

PURPOSE

FireFly addresses the lack of real-time wildfire monitoring in remote, high-risk areas where fixed sensors and cameras may not exist. Its purpose is to give responders earlier, safer awareness of possible ignition points by collecting aerial visual and telemetry data, locating detected fires, and sending actionable alerts to a ground station before conditions escalate.

SOLUTION

FireFly combines ESP32-based drone telemetry, computer-vision fire detection, and a live ground-station map. Drone nodes report GPS position, altitude, and fire-detection status. A YOLO-based vision pipeline detects sustained fire or smoke in video. When multiple drones observe the same event, the ground station triangulates their bearings to estimate the fire's GPS coordinates and displays drone positions, fire markers, and alerts for the operator.

OBJECTIVES

- Detect sustained fire or smoke from aerial video
- Send drone GPS and fire-status telemetry to a ground station
- Triangulate fire location from 2+ drone observations
- Display live drone positions, fire alerts, and estimated fire coordinates on a map
- Validate the end-to-end pipeline through simulation and unit tests

TARGET USERS

- Wildfire operations coordinators who need fast, clear situational awareness
- Fire-response and forestry field crews operating in dangerous or hard-to-access terrain
- Environmental researchers collecting fire-risk and field telemetry data
- Future engineering teams extending the prototype into full flight hardware and field deployment

SOFTWARE

Ground Station

- Flask based telemetry server
- Live interactive map interface
- Real time drone tracking and fire visualization

Vision System

- YOLOv8 fire/smoke detection pipeline
- OpenCV video processing
- Sustained detection filtering to reduce false positives

Localization

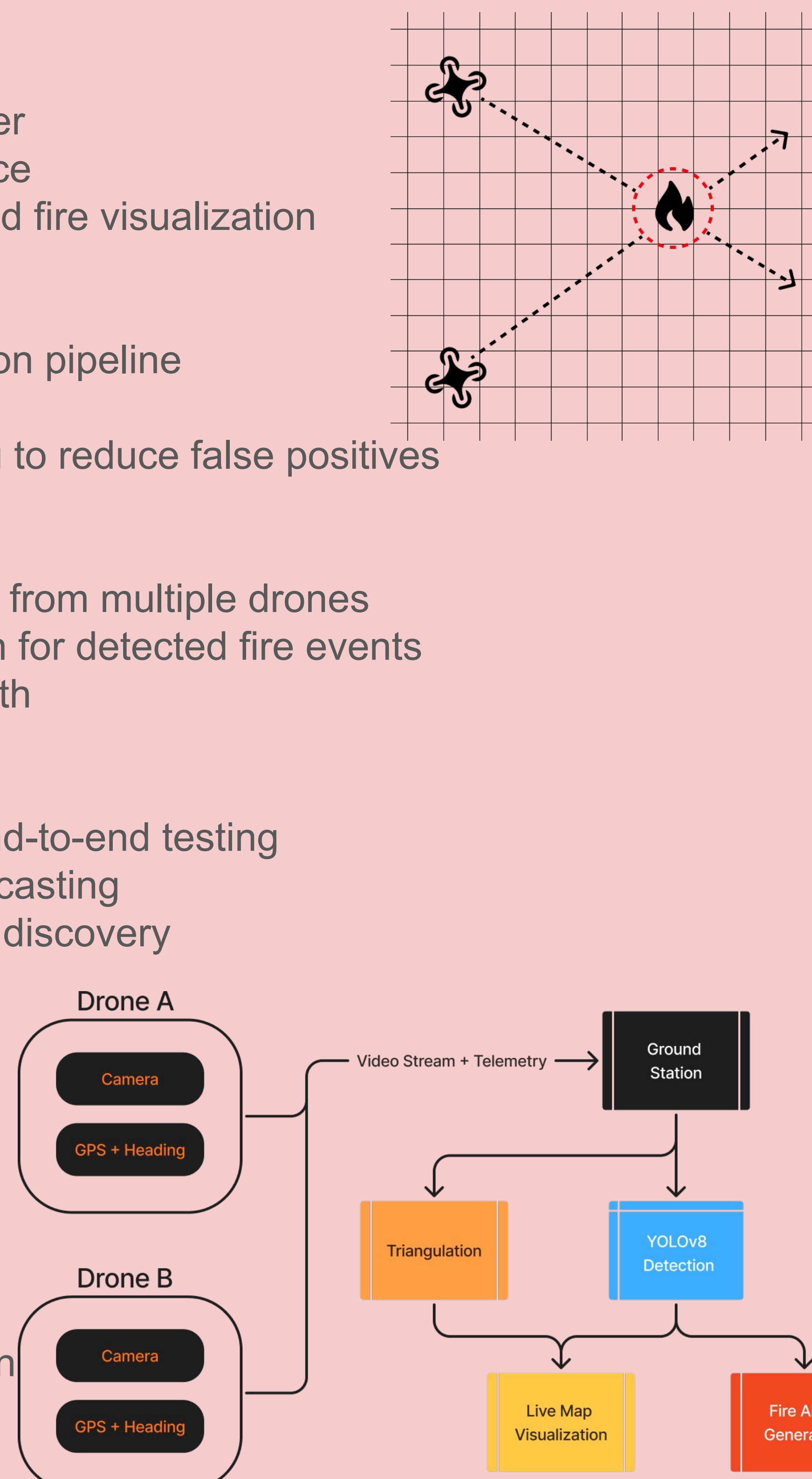
- Bearing based triangulation from multiple drones
- GPS coordinates estimation for detected fire events
- Unit tested triangulation math

Simulation + Testing

- Simulated drone fleet for end-to-end testing
- Automated telemetry broadcasting
- Local network camera auto discovery

Technologies Used

- Python
- Flask
- OpenCV
- YOLOv8
- ESP-IDF/FreeRTOS
- ESP-NOW
- JavaScript map visualization



HARDWARE

Drone Platform

- ESP32-based flight/telemetry nodes
- Camera equipped aerial platform
- GPS module for positional telemetry
- Wireless communication

Core Components

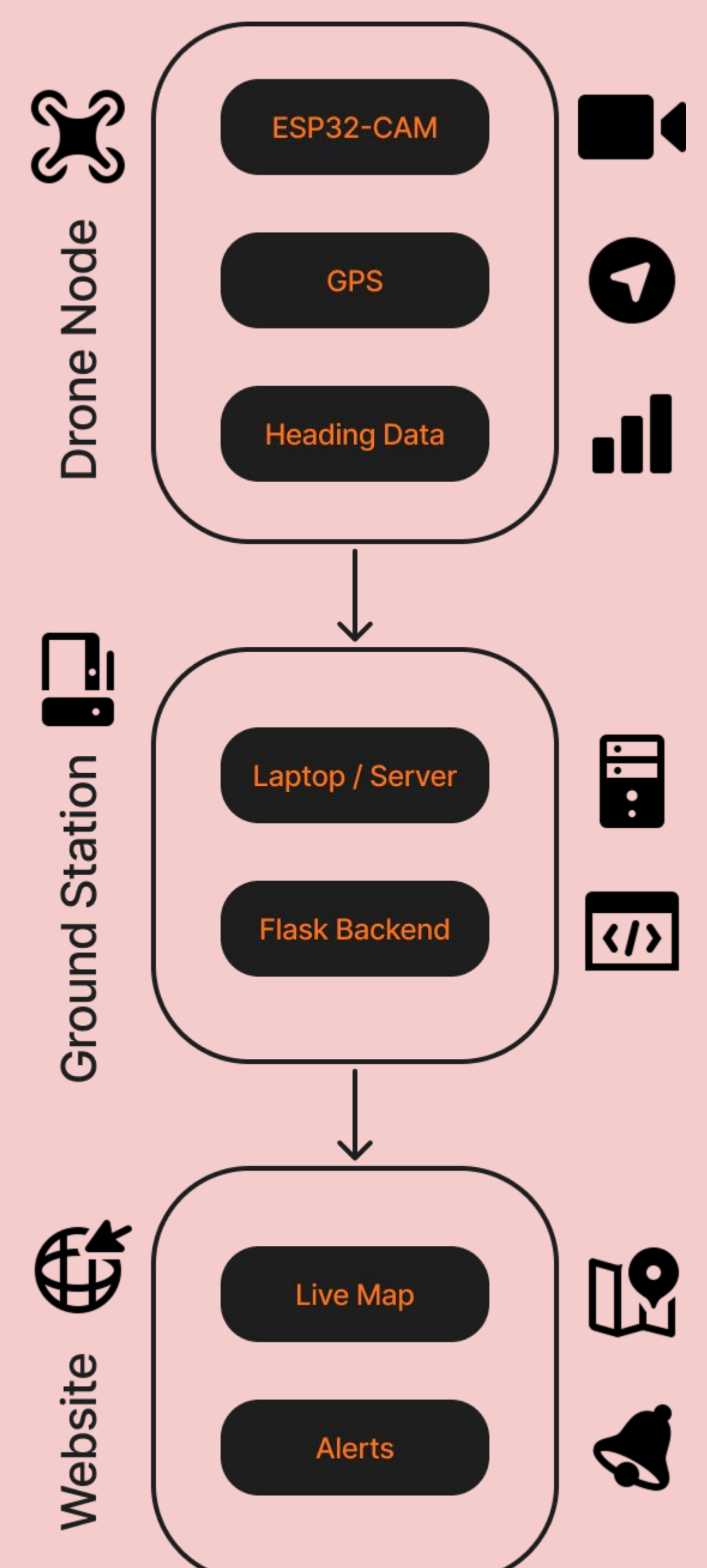
- ESP32 microcontroller
- ESP32-CAM video module
- GPS receiver

Communication

- ESP-NOW broadcast telemetry between drone nodes
- WiFi link to ground station
- Real time GPS and fire status transmission

Sensors/Data Sources

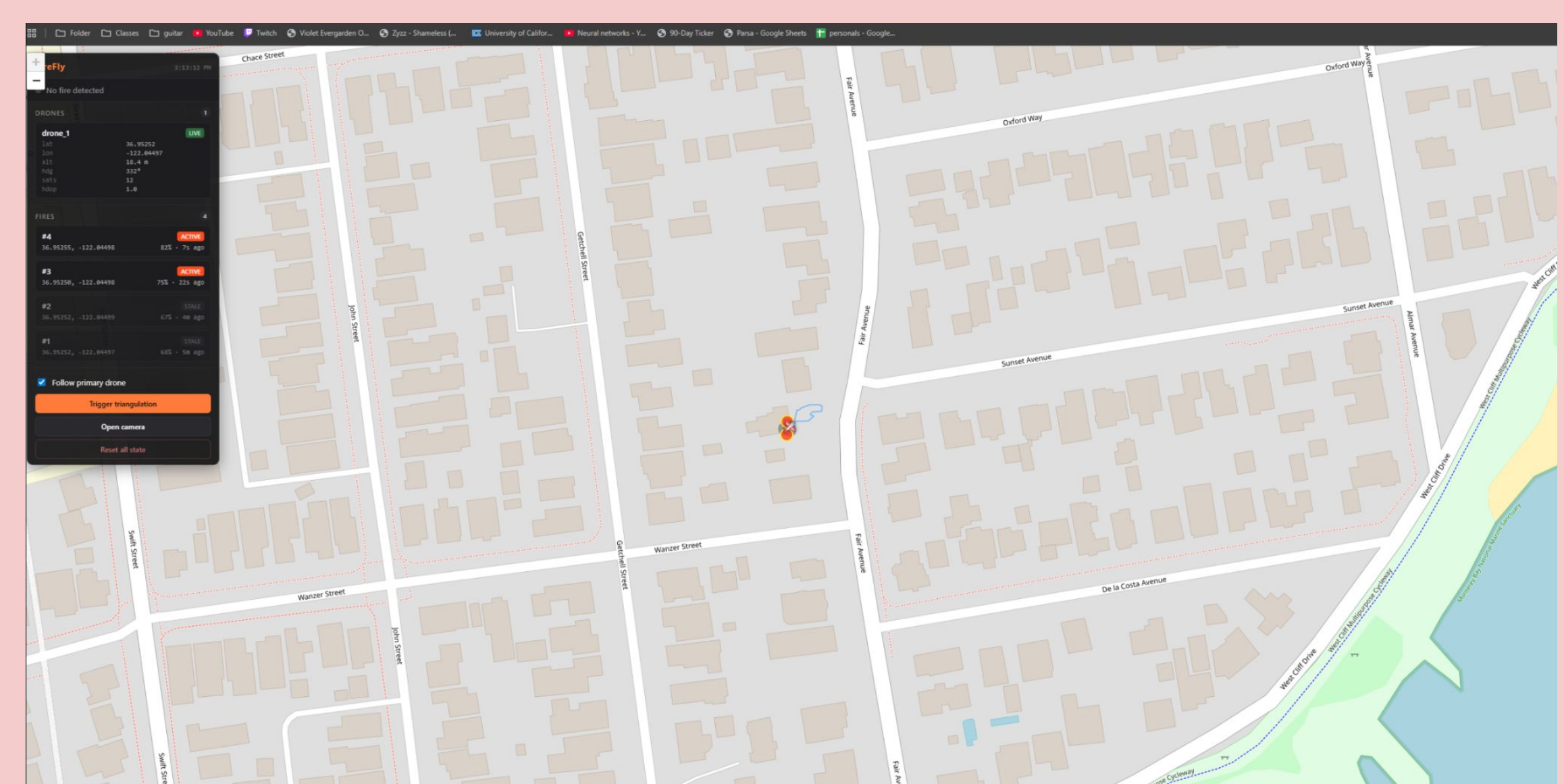
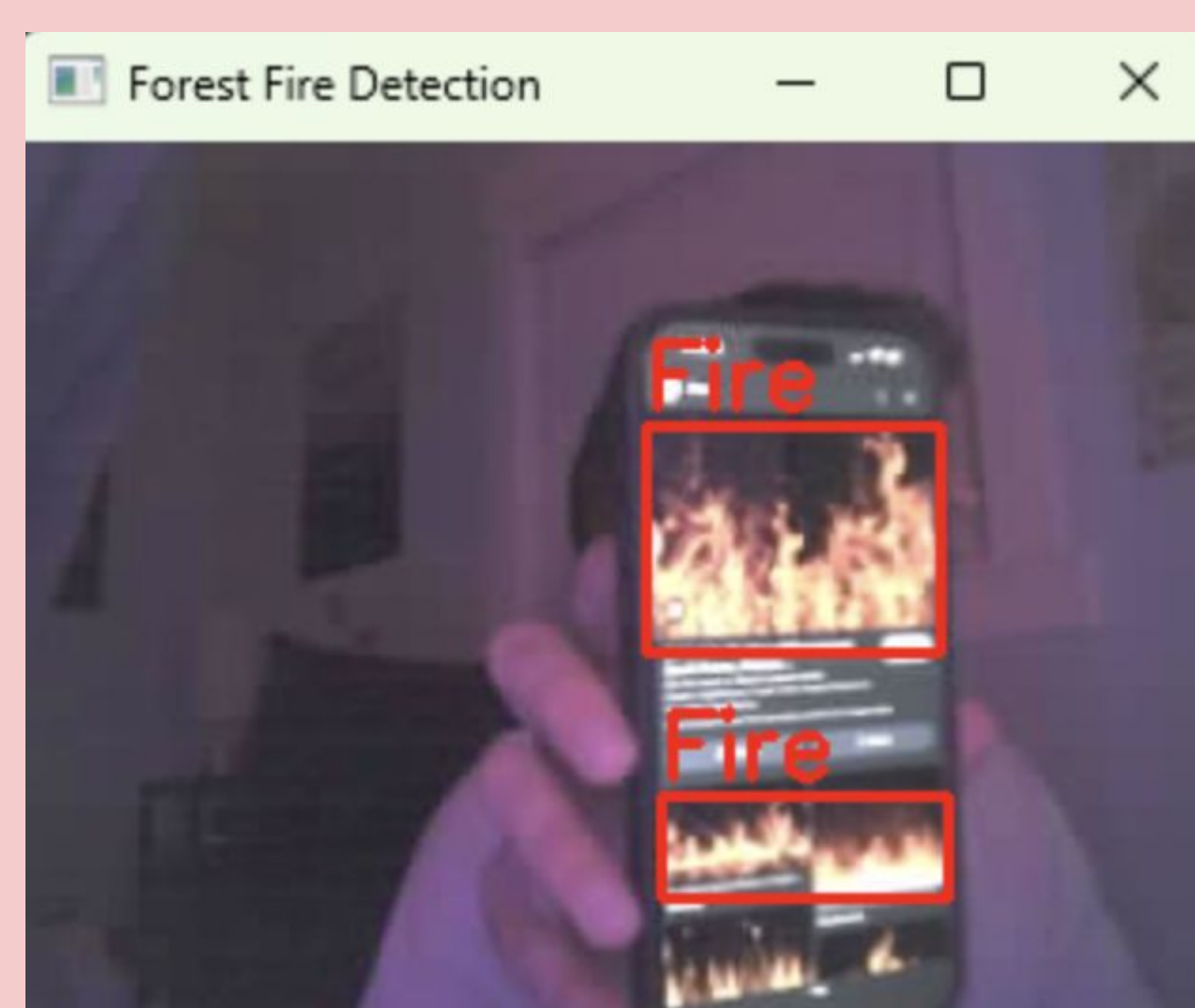
- Camera stream for fire detection
- GPS coordinates
- Drone heading/bearing data



REFERENCES

- [1] B. Yun et al., "FFYOLO: A Lightweight Forest Fire Detection Model Based on YOLOv8," Fire, 7(3), 93, 2024. doi:10.3390/fire7030093
- [2] Espressif Systems, "ESP-NOW Programming Guide," docs.espressif.com.
- [3] Ultralytics, "YOLO Documentation," docs.ultralytics.com; fire model adapted from thilak-r/Forest-fire-detection-using-YOLOv8.
- [4] Leaflet, "JavaScript Library for Interactive Maps," leafletjs.com. Map data: © OpenStreetMap contributors.

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Author Name

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Section Headings

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References, Acknowledgments

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Captions (description of figures - typically below figures)

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